

L1 — Equivalence Completion (Paper 10): Absence of Tree-Level Flavor-Changing Neutral Currents under Admissible Redescription

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0. Scope and Non-Negotiables

This paper is eliminative. It introduces no new primitives, no new ontology, no new anchors, and no new tensor load. All results are invariant under admissible redescription.

We work with fixed fermion content, fixed gauge representations, and a single electroweak Higgs doublet. No additional Higgs multiplets, no compensating fermions, and no non-gauge-invariant patches are permitted.

1. Apparent Freedom of Neutral Currents

In broken-phase bookkeeping, neutral current interactions appear sensitive to fermion mixing structure. Absent further constraints, this suggests the possibility of tree-level flavor-changing neutral currents (FCNCs).

In gauge-phase bookkeeping, neutral currents arise from fixed gauge generators acting diagonally in representation space. This creates an apparent tension between the two skins.

2. Admissibility Envelope

Allowed redescriptions include:

- unitary field redefinitions preserving gauge-covariant kinetic terms,
- basis changes within fixed fermion multiplets,
- admissible transport between gauge-phase and broken-phase skins.

Any structure that generates non-transportable interaction terms is inadmissible.

3. Transport of Neutral Currents

Neutral current interactions are fixed by gauge generators in the gauge-phase skin. Transport to the broken-phase skin proceeds via admissible diagonalization of the Yukawa operators.

Because the same unitary transformations that diagonalize the mass matrices also act on the fermion fields, neutral current couplings remain diagonal provided the underlying gauge structure is unchanged.

4. Failure of Tree-Level FCNC under Fixed Load

Assume the existence of tree-level FCNCs in the broken-phase skin. Transport back to the gauge-phase skin.

Such terms cannot be represented without: (i) introducing additional Higgs multiplets coupling differently to fermion families, (ii) adding new fermion representations, or (iii) modifying gauge generators. Each option violates the declared load or admissibility constraints.

Therefore, tree-level FCNCs are inadmissible under fixed fermion content and a single Higgs doublet.

5. Core Result

Theorem 1 (Absence of Tree-Level FCNC under Admissible Transport). *Fix the fermion content, gauge representations, and a single electroweak Higgs doublet. Under admissible redescription and global constraints (no new load, no patches), neutral current interactions are flavor-diagonal at tree level.*

Tree-level flavor-changing neutral currents are inadmissible.

Corollary 1 (GIM Structure as Transport Necessity). *The Glashow–Iliopoulos–Maiani cancellation structure is not an additional assumption, but a transport necessity under admissible redescription with fixed declared load.*

Corollary 2 (Skin Illusion of FCNC Freedom). *Any apparent freedom to introduce tree-level FCNCs is a bookkeeping artifact of broken-phase representation. Cross-skin admissibility collapses this illusion.*

6. Closure Statement

The absence of tree-level flavor-changing neutral currents is enforced by admissible transport under fixed fermion content and Higgs structure. No additional assumptions are required.